

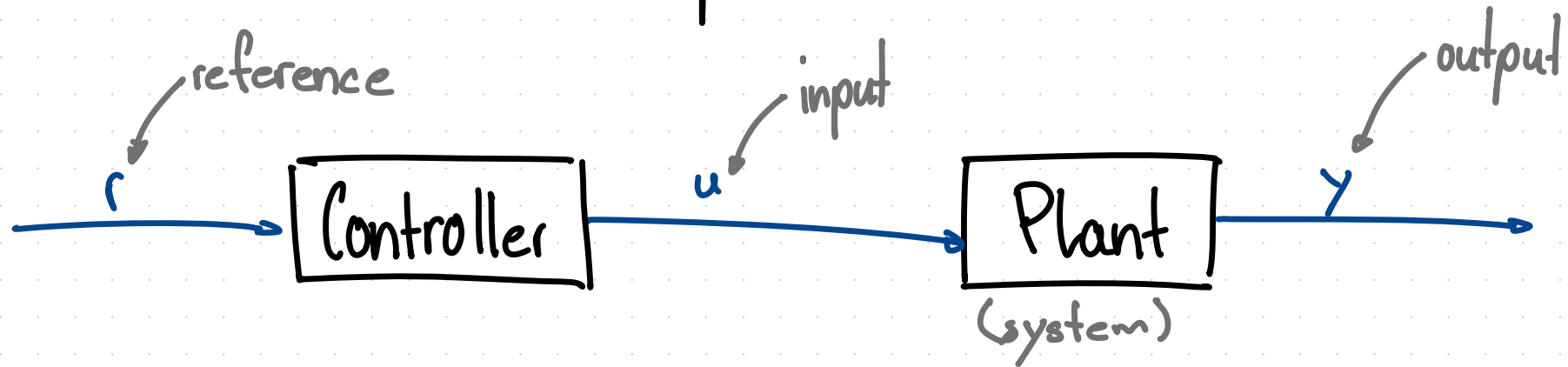
Exercise Session 3

What we're doing today:

1. Introduction / Orientation
2. System Classification
3. Linearization
4. Equilibria
5. Example
6. Quiz

Where we left off...

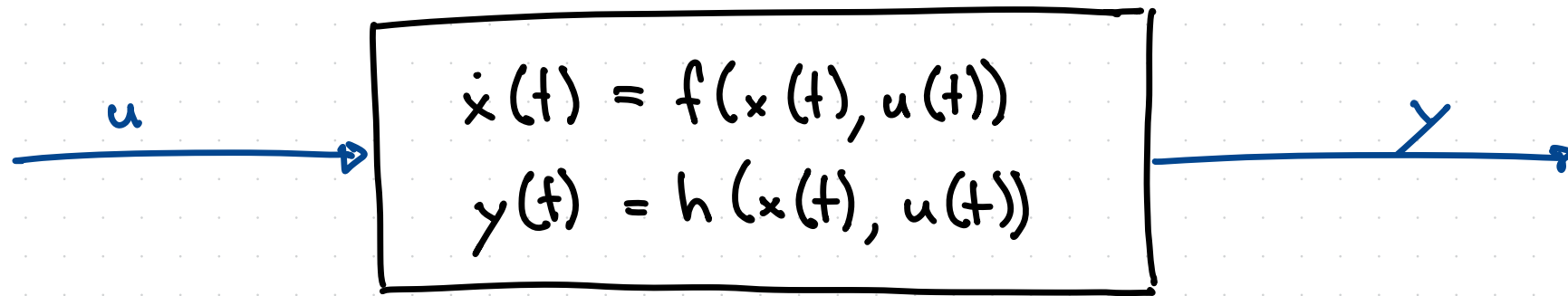
- General structure of our problem:



- To solve our problem we need to understand our system:



- Which is really just...



- ... and we can find $f(x(t), u(t))$ & $h(x(t), u(t))$ through modeling.

2. System Classification

We can classify our System in 4 different ways:

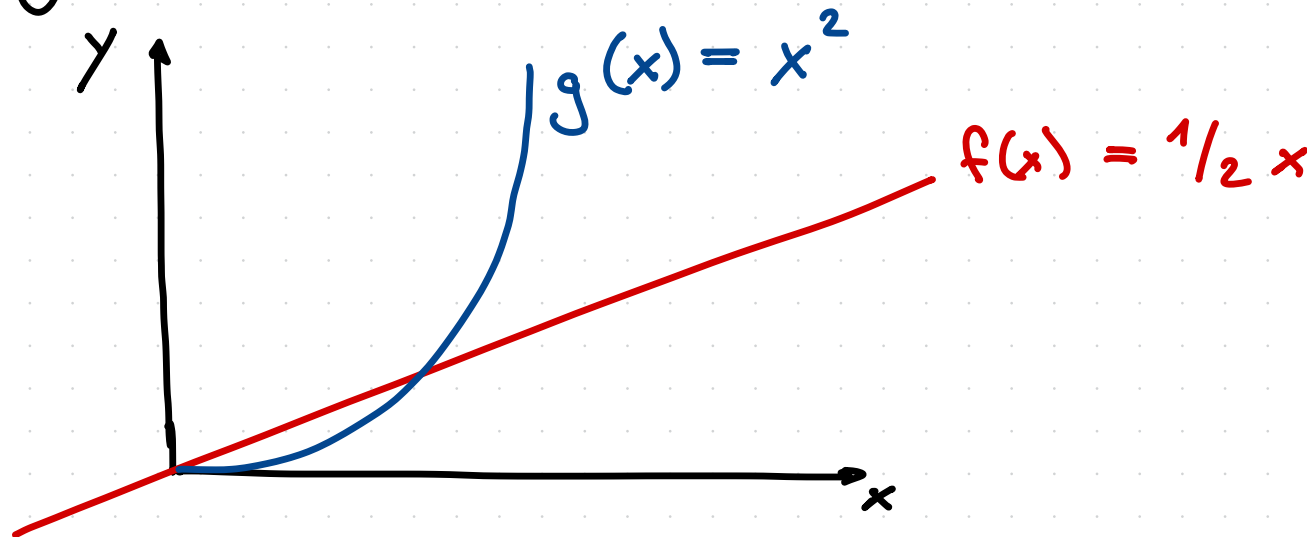
i) Linearity

Def: A system is **linear** if it is ...

... **Additive**: $\Sigma(u_1 + u_2) = \Sigma(u_1) + \Sigma(u_2)$

... **Homogeneous**: $\Sigma(\alpha u) = \alpha \Sigma(u)$

Intuition:



.... Why do we care? **Superposition!**

$$y = \Sigma(u)$$

$$\rightarrow u = \alpha u_1 + \alpha u_2$$

$$\begin{aligned} \Rightarrow y &= \Sigma(u) = \Sigma(\alpha u_1 + \alpha u_2) = \alpha \Sigma(u_1) + \alpha \Sigma(u_2) \\ &:= y_1 + y_2 \end{aligned}$$

ii) Causality

Def: A system is causal if $\forall t, y(t)$ only depends on $(-\infty, t]$.

A system is strictly causal if $\forall t, y(t)$ only depends on $(-\infty, t)$.

Ex:

Causal:

- $y(t) = 3u(t)$
- $y(t) = t \cdot \cos(t-3)$

VS.

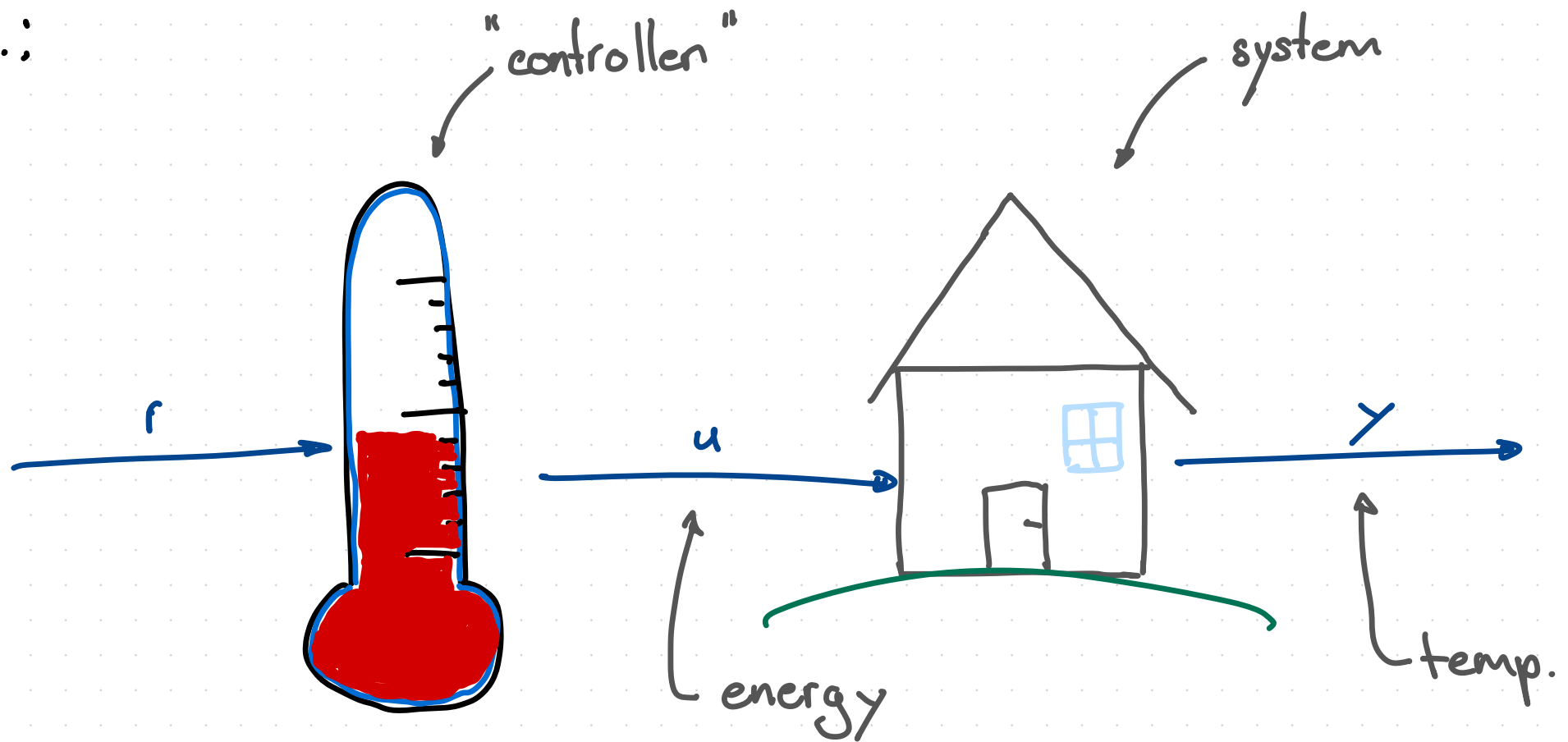
Non-causal:

- $y(t) = u(2t)$
- $y(t) = \int_{-\infty}^{t+1} u(\tau) d\tau$
- $y(t) = (t+2) \cos(u(t))$

Intuition:

All practically realizable systems are causal!
(impossible to implement a non-causal system in "the real world")

Ex.:



iii. Memoryless (static) vs. Dynamic Systems

Def.: A system is **memoryless (static)** if $y(t)$ only depends on $u(t)$ $\rightarrow$ output at present time only depends on input at present time.

Ex.:

Static:

- $y(t) = 3u(t)$
- $y(t) = 2^{-(t+1)} u(t)$
- $y(t) = \sqrt{\sin(u^2(t))}$

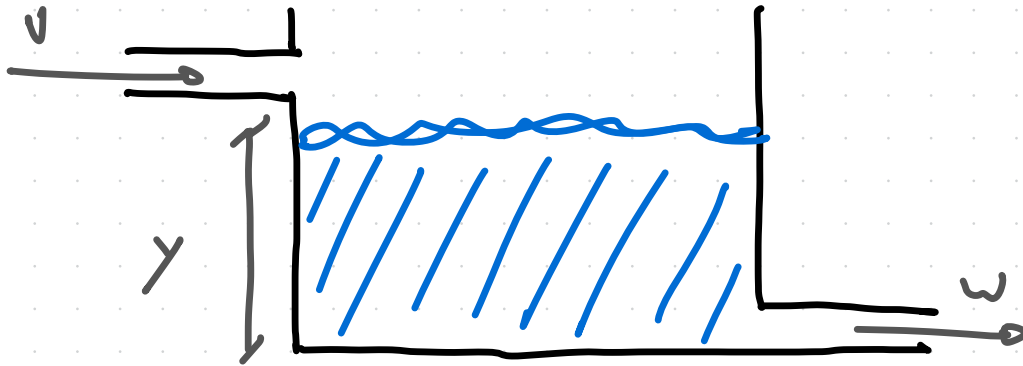
vs.

Dynamic:

- $y(t) = \int_{-\infty}^t u(\tau) d\tau$
- $y(t) = \dot{u}(t)$
- $y(t) = u(2t)$

Rule-of-thumb: Integral / Derivative $\Rightarrow$ Dynamic!

Intuition:



$$u = v - w$$

$$\rightarrow y(t) = \int_{-\infty}^t u(\tau) d\tau$$

$\Rightarrow$ tank acts like an integral!

$\Rightarrow$ DYNAMIC

iv) Time-variance:

Def: A system is time-invariant, if the same input produces the same output at any time.

Ex.:

Time-invariant:

$$\cdot y(t) = \cos(u(t))$$

$$\cdot y(t) = 8u(t)$$

vs.

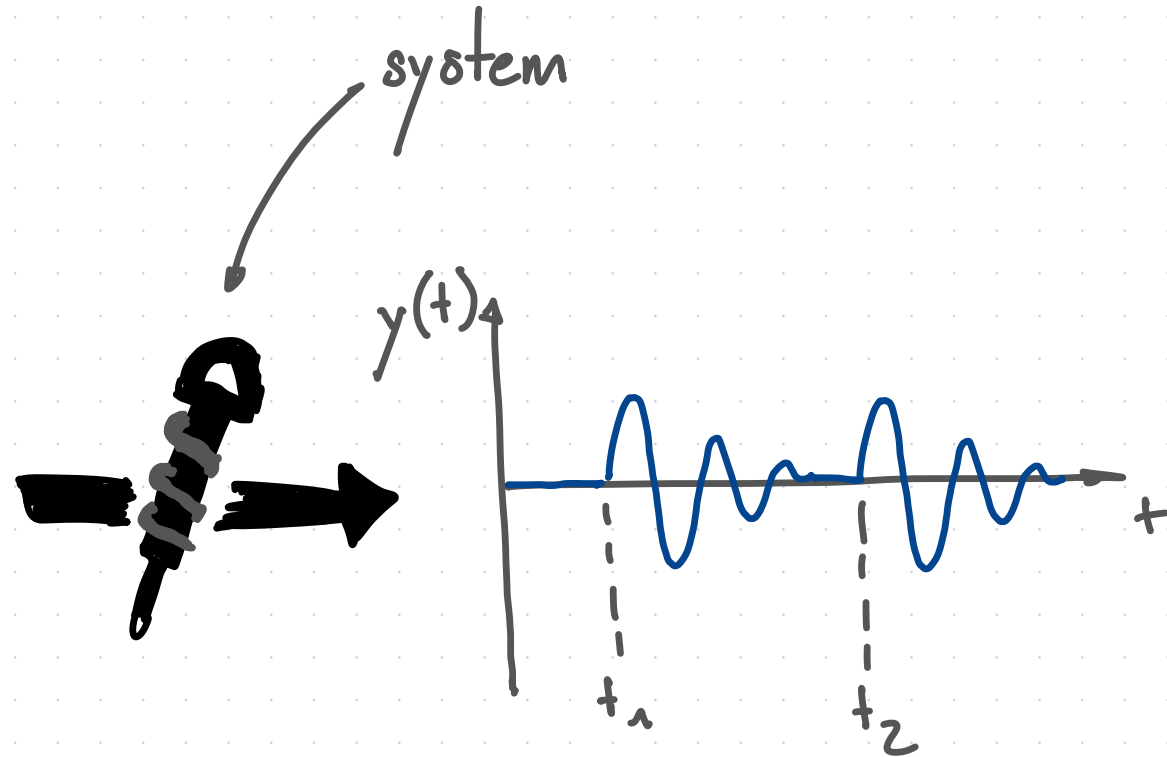
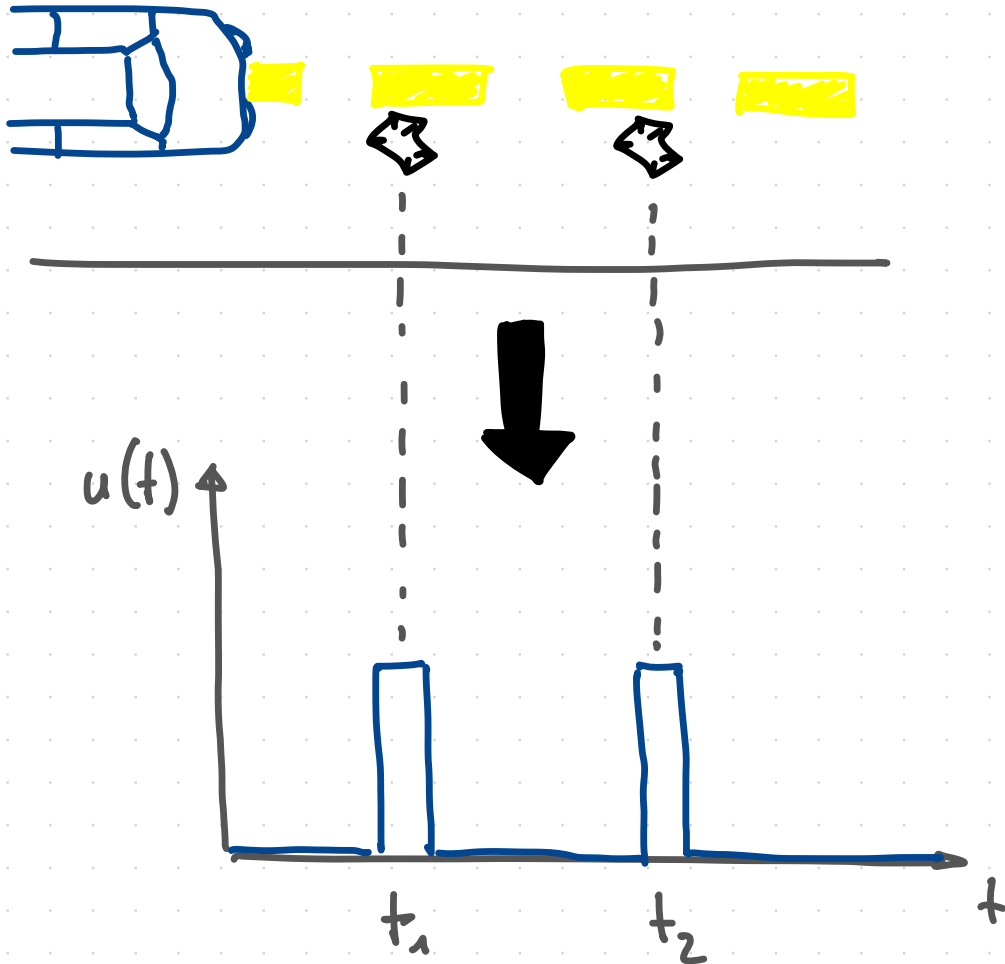
Time-variant:

$$\cdot y(t) = 3t \cos(t)$$

$$\cdot y(t) = \cos(t) \cdot u(t)$$

Intuition

Car-damping system:



Let's test our knowledge:

Q1)

$$y(t) = \sqrt{t - \cos^2(t)} \cdot u(t^2)$$

Statement	True	False
The system is linear	x	
The system is static		x
The system is time-invariant		x

Q2)

$$u(t) \in \mathbb{R}^2$$

$$y(t) \in \mathbb{R}$$

$$x(t) = u_1(t - T), \quad \text{mit der Konstante } T = 1 \text{ s}$$

$$y(t) = -x^2(t) + 2 \cdot u_2(t)$$

Behauptung	wahr	falsch
Das System ist linear		x
Das System ist zeit-invariant	x	
Das System ist dynamisch	x	
Das System ist SISO		x

Q3)

$$x_1(t) = \int_{-\infty}^t u(\tau - 3) d\tau$$

$$x_2(t) = u(t)$$

$$y(t) = x_1(t) + x_2(t)$$

Statement	True	False
The system is linear	x	
The system is static		x
The system is time-invariant	x	

3. Linearization

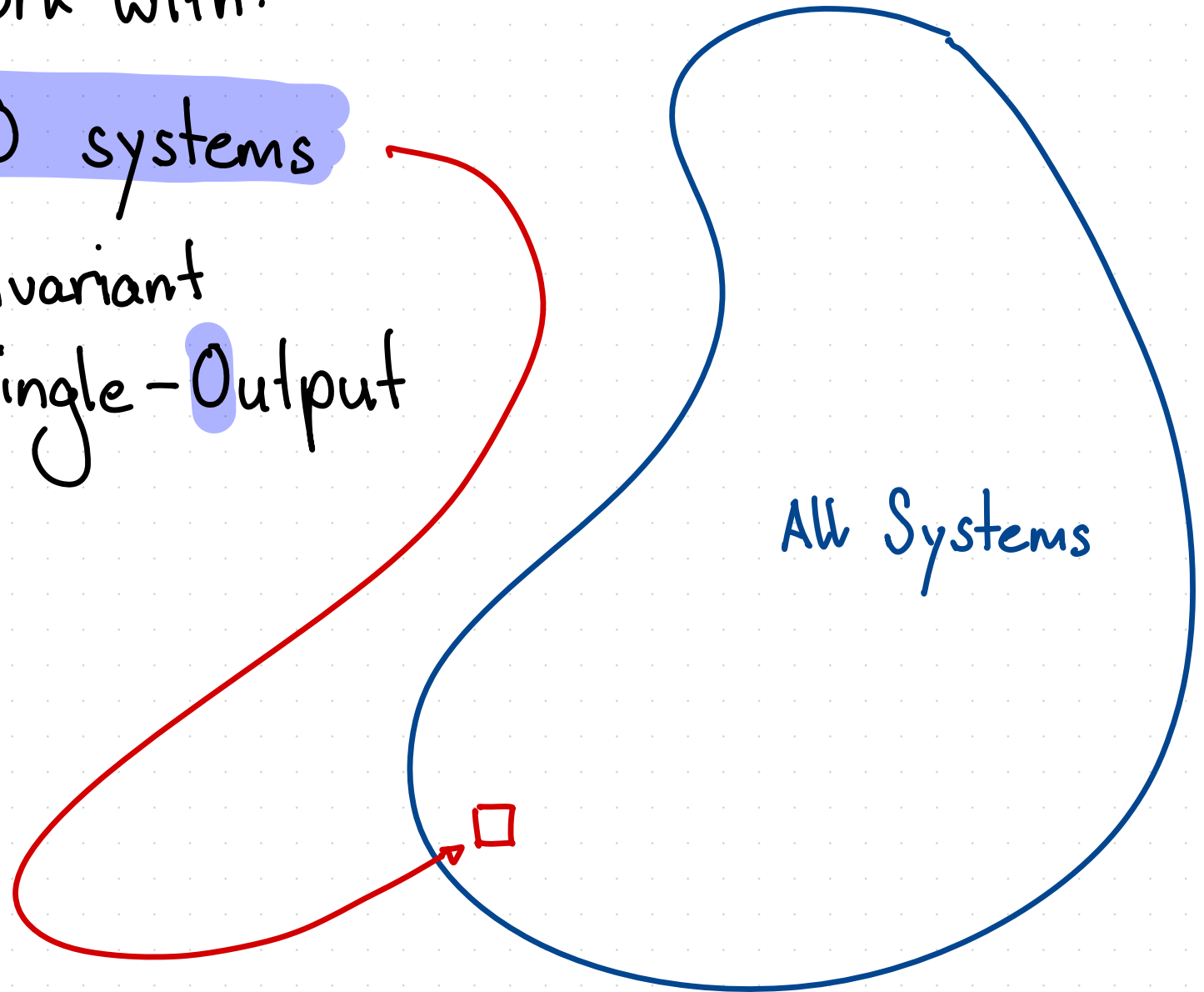
Motivation...

3. Linearization

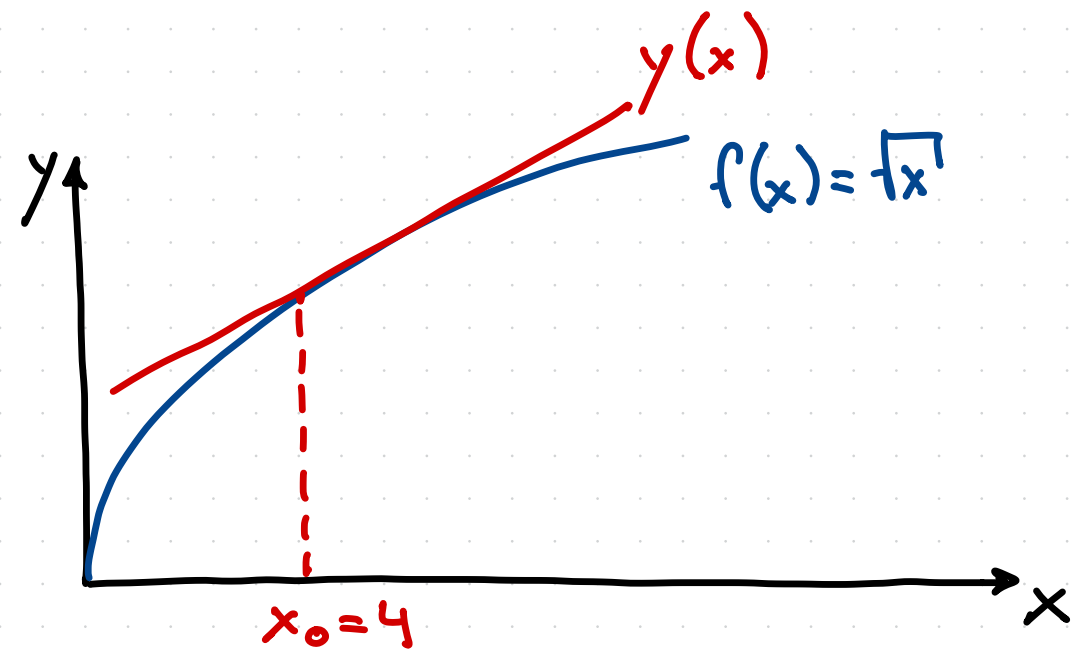
In CSI we will work with:

LTI - SISO systems

↳ Linear, Time-invariant
Single-Input-Single-Output



Linearization Warm-up:



→ We can approximate $f(x)$ at any point x_0 with a tangent fct.
↳ Let's choose $x_0 = 4$

General formula of a straight line:

$$y = mx + b \quad \begin{cases} m = \frac{df(x)}{dx} = \frac{1}{2\sqrt{x}} \\ b = f(x) \end{cases}$$

⇒ Linearization at x_0 :

$$\underline{\underline{y(x) = \underbrace{\frac{1}{2\sqrt{x_0}}}_{\text{slope}} \underbrace{(x - x_0)}_{\text{shift by } x_0} + \underbrace{\sqrt{x_0}}_{\text{offset}}}}$$

Let's generalize this:

• General structure of our system:

$$\begin{array}{l} \dot{x}(t) = f(x(t), u(t)) \quad (\text{I}) \\ y(t) = h(x(t), u(t)) \quad (\text{II}) \end{array}$$

• To linearize, we use the Taylor series:

$$T(f(x)) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x)}{n!} \Big|_{x=x_0} \cdot (x-x_0)^n$$

• Apply to (I) & (II):

$$\dot{x}(t) = f(x_0, u_0) + \left. \frac{\partial f(x, u)}{\partial x} \right|_{(x_0, u_0)} \cdot (x - x_0) + \left. \frac{\partial f(x, u)}{\partial u} \right|_{(x_0, u_0)} \cdot (u - u_0) + \text{h.o.t.}$$

$$\dot{y}(t) = h(x_0, u_0) + \left. \frac{\partial h(x, u)}{\partial x} \right|_{(x_0, u_0)} \cdot (x - x_0) + \left. \frac{\partial h(x, u)}{\partial u} \right|_{(x_0, u_0)} \cdot (u - u_0) + \text{h.o.t.}$$

→ neglect the h.o.t

→ choose linearization point at an equilibrium point

$$(x_0, u_0) = (x_e, u_e), \text{ where: } \dot{x}(t) = f(x_e, u_e) = 0$$

... Which leaves us with:

$$x(t) = \left. \frac{\partial f(x,u)}{\partial x} \right|_{(x_e, u_e)} \cdot (x - x_e) + \left. \frac{\partial f(x,u)}{\partial u} \right|_{(x_e, u_e)} \cdot (u - u_e)$$

$$y(t) = h(x_e, u_e) + \left. \frac{\partial h(x,u)}{\partial x} \right|_{(x_e, u_e)} \cdot (x - x_e) + \left. \frac{\partial h(x,u)}{\partial u} \right|_{(x_e, u_e)} \cdot (u - u_e)$$

→ do a substitution:

$$\cdot \Delta x = x - x_e \rightarrow \Delta \dot{x} = \dot{x}$$

$$\cdot \Delta y = y - y_e = y - h(x_e, u_e)$$

$$\Delta \dot{x} = \left. \frac{\partial f(x,u)}{\partial x} \right|_{(x_e, u_e)} \Delta x + \left. \frac{\partial f(x,u)}{\partial u} \right|_{(x_e, u_e)} \Delta u$$

⇒

$$\Delta y = \left. \frac{\partial h(x,u)}{\partial x} \right|_{(x_e, u_e)} \Delta y + \left. \frac{\partial h(x,u)}{\partial u} \right|_{(x_e, u_e)} \Delta u$$

→ rename out of convention:

$$\cdot \Delta x = x \quad \cdot \frac{\partial f}{\partial x} \Big|_{(x_e, u_e)} = A \quad \cdot \frac{\partial f}{\partial u} \Big|_{(x_e, u_e)} = B$$

$$\cdot \Delta y = y \quad \cdot \frac{\partial h}{\partial x} \Big|_{(x_e, u_e)} = C \quad \cdot \frac{\partial h}{\partial u} \Big|_{(x_e, u_e)} = D$$

⇒ We arrive at the standard form for a linearized sys:

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t)$$

$$x(0) = x_0$$

→ for the general case with multiple x_i & u_i , we must evaluate the **Jacobi-matrices**:

$$A = \frac{\partial f(x, u)}{\partial x} \bigg|_{(x_e, u_e)} = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_n} \end{pmatrix} \bigg|_{(x_e, u_e)} \in \mathbb{R}^{n \times n}$$

$$B = \frac{\partial f(x, u)}{\partial u} \bigg|_{(x_e, u_e)} = \begin{pmatrix} \frac{\partial f_1}{\partial u_1} & \dots & \frac{\partial f_1}{\partial u_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial u_1} & \dots & \frac{\partial f_n}{\partial u_m} \end{pmatrix} \bigg|_{(x_e, u_e)} \in \mathbb{R}^{n \times m}$$

$$C = \frac{\partial h(x, u)}{\partial x} \bigg|_{(x_e, u_e)} = \begin{pmatrix} \frac{\partial h_1}{\partial x_1} & \dots & \frac{\partial h_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_p}{\partial x_1} & \dots & \frac{\partial h_p}{\partial x_n} \end{pmatrix} \bigg|_{(x_e, u_e)} \in \mathbb{R}^{p \times n}$$

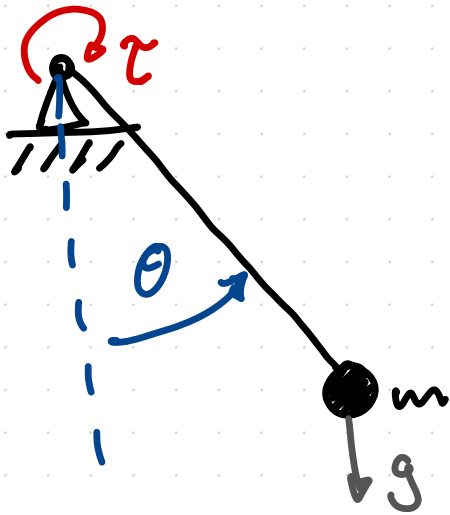
$$D = \frac{\partial h(x, u)}{\partial u} \bigg|_{(x_e, u_e)} = \begin{pmatrix} \frac{\partial h_1}{\partial u_1} & \dots & \frac{\partial h_1}{\partial u_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_p}{\partial u_1} & \dots & \frac{\partial h_p}{\partial u_m} \end{pmatrix} \bigg|_{(x_e, u_e)} \in \mathbb{R}^{p \times m}$$

4. Equilibria

• What we used so far:

$$\dot{x} = f(x_e, u_e) = 0$$

• Ex.: Pendulum



→ state: $x = \begin{pmatrix} \dot{\theta} \\ \theta \end{pmatrix}$

→ input: τ

When to use
linearization
Operation
point...

→ for which states can we do $\dot{\theta} = 0$? → $\forall \theta$

→ for which states with zero input $\tau = 0$ does $\dot{\theta} = 0$ hold?

$$x_{e1} = (0, 0)$$

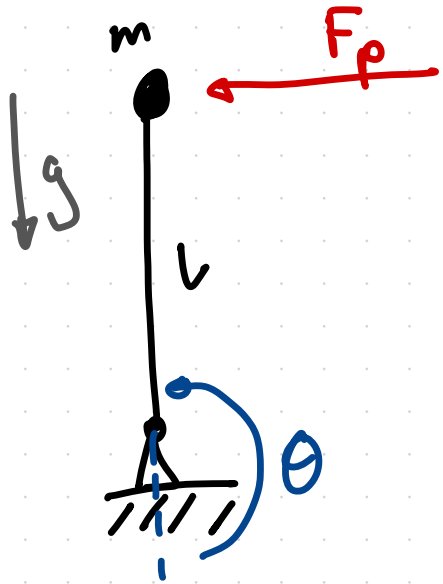
stable ↗

$$x_{e2} = (\pi, 0)$$

↖ unstable

5. Example

Pendulum Linearization:



$$\cdot u(t) = F_p(t)$$

$$\cdot x_1(t) = \theta(t)$$

$$\cdot x_2(t) = \dot{x}_1(t) = \dot{\theta}(t)$$

$$\cdot y(t) = \theta(t)$$

$$\rightarrow x(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} \Rightarrow \dot{x}(t) = \begin{pmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{pmatrix} = \begin{pmatrix} x_2(t) \\ -\frac{g}{l} \sin(x_1(t)) + \frac{1}{ml} \cos(x_1(t)) u(t) \end{pmatrix}$$

$$\rightarrow y(t) = x_1(t)$$

$\Rightarrow$ Change of tactics!

$$\left. \begin{array}{l} \dot{x}(t) = f(x(t), u(t)) \\ y(t) = h(x(t), u(t)) \end{array} \right\} \begin{array}{l} \dot{x}(t) = Ax(t) + Bu(t) \\ y(t) = Cx(t) + Du(t) \end{array}$$

1) Choose desired operation point: $x_e = \begin{pmatrix} \pi \\ 0 \end{pmatrix}$, $u_e = 0$

2) Evaluate Jacobi matrices:

$$A = \left(\begin{array}{cc} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{array} \right) \bigg|_{(\pi, 0, 0)} = \left(\begin{array}{cc} 0 & 1 \\ -\frac{g}{l} \cos(x_1) - \frac{1}{mb} \sin(x_1) u_1 & 0 \end{array} \right) \bigg|_{(\pi, 0, 0)} = \left(\begin{array}{cc} 0 & 1 \\ g/l & 0 \end{array} \right)$$

$$\cdot B = \begin{pmatrix} \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial u} \end{pmatrix} \bigg|_{(\pi, 0, 0)} = \begin{pmatrix} 0 \\ \frac{1}{mb} \cos(x_1) \end{pmatrix} \bigg|_{(\pi, 0, 0)} = \begin{pmatrix} 0 \\ -\frac{1}{mb} \end{pmatrix}$$

$$\cdot C = \begin{pmatrix} \frac{\partial h}{\partial x_1} & \frac{\partial h}{\partial x_2} \end{pmatrix} \bigg|_{(\pi, 0, 0)} = \begin{pmatrix} 1 & 0 \end{pmatrix} \bigg|_{(\pi, 0, 0)} = \begin{pmatrix} 1 & 0 \end{pmatrix}$$

$$\cdot D = \frac{\partial h}{\partial u} \bigg|_{(\pi, 0, 0)} = 0 \bigg|_{(\pi, 0, 0)} = 0$$

$$\Rightarrow \dot{x}(t) = \begin{pmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ g/b & 0 \end{pmatrix} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} + \begin{pmatrix} 0 \\ -\frac{1}{mb} \end{pmatrix} u(t) //$$

$$y(t) = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}$$

6. QUIZ

Q4: For a time-invariant system we can generalize the initial condition as follows: $x(t_0) = x_0 \Rightarrow x(0) = x_0$

☒ TRUE

☐ FALSE

Q5: The following system can be strictly causal:

$$\dot{x}(t) = Ax + Bu$$

$$y(t) = Cx + Du$$

with $A, B, C, D \neq 0$

☐ TRUE

☒ FALSE

Q6: These three terms mean the same:

Linearization point / Equilibrium point / operation point

☐ TRUE

☒ FALSE